

Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

POSTLAB EXERCISES (18 points)*Submit the postlab to the TA at the end of the lab.***Part 1: Absolute Voltage Measurement (10 points)****Question 3** **$\frac{1}{2}$ point**

Record your displacement measurements for each applied voltage in Data Table 1.

Data table 1: *Displacement vs. applied voltage.*

Voltage [V]	Displacement [mm]	<i>s</i>
0		
1000		
2000		
2500		
3000		

Question 4**2 points**

Make a plot of displacement s versus V^2 using the data from Data Table 1. Plot V^2 on the horizontal axis and s on the vertical axis. Draw a best-fit straight line, measure its slope, and show your calculation. Include axis labels with units.

Question 5 **$\frac{1}{2}$ point**

Record your repeated displacement measurements for the unknown voltage in Data Table 2, and compute the average displacement s_{avg} .

Data table 2: *Displacement measurements for the unknown voltage.*

Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	s_{avg} [mm]

Unknown voltage (from power supply): $V_{\text{known}} =$ _____

Balancing force (null method): $F_{\text{null}} =$ _____

Question 6**1 point**

Using the slope found in the graph above (in mm V^{-2}) and s_{avg} from Data Table 2, calculate the unknown voltage. Compute the percent error using the known value.

Question 7**1 point**

Using eq. (6) and the balancing force F_{null} , calculate the unknown voltage a second time.

Question 8**4 points**

What is the limiting experimental parameter for measuring an unknown voltage in each method? That is, what variable has the greatest uncertainty?

(a) (1 point) Displacement method:

(b) (1 point) Null method:

Question 9**1 point**

Which method of measurement is more reliable? Why? Be specific.

Part 2: Electric Field Mapping (8 points)

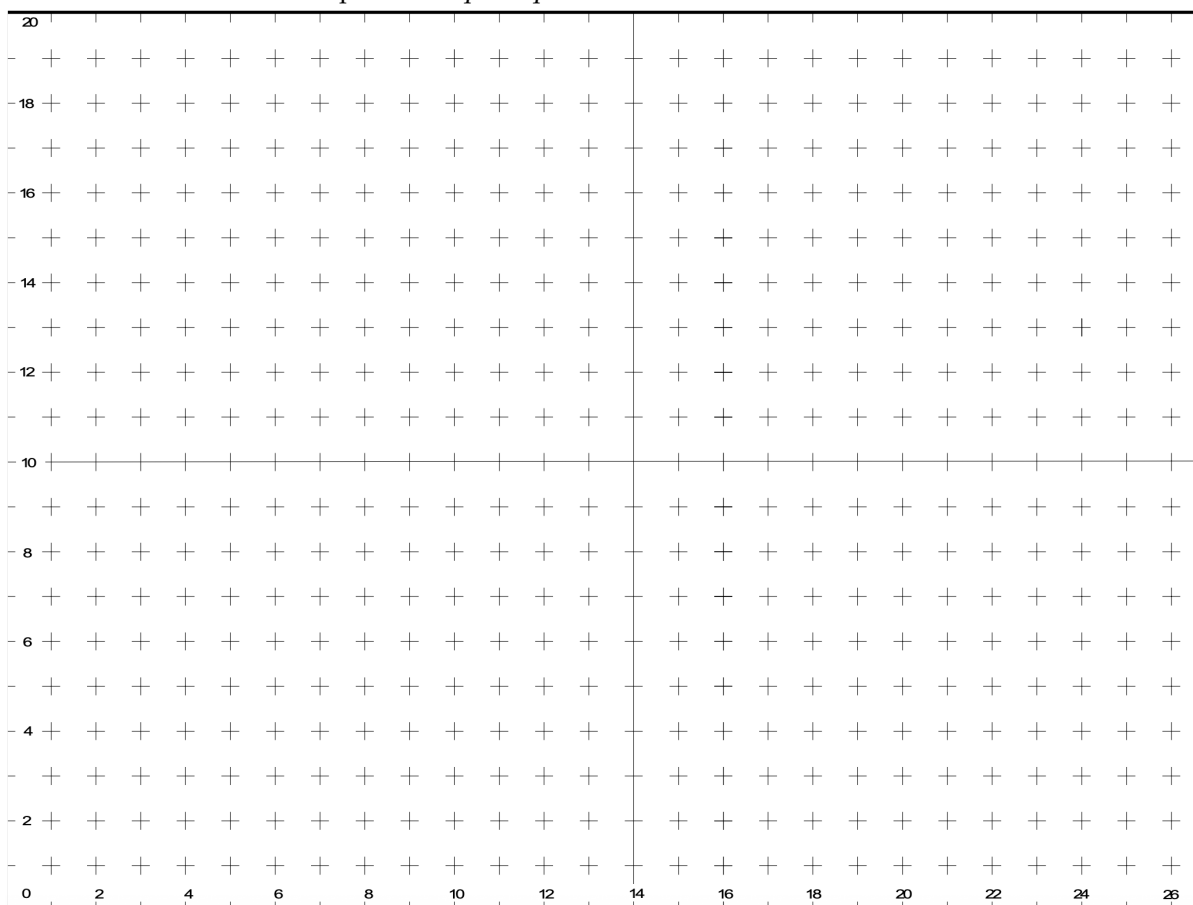
Question 10**1 point**

Draw the electric field lines you would expect for an isolated positive point charge $+q$ below. Identify two arbitrary regions and label which has the stronger field. Explain how you can determine this from the field line pattern.

Question 11**3 points**

In Graph 1, sketch the electric field lines according to your measured mapping of equipotential surfaces for the two-point electrode arrangement. Include arrows showing the direction of the field lines.

Graph 1: *Two-point potential and electric field lines.*



+1+1+1

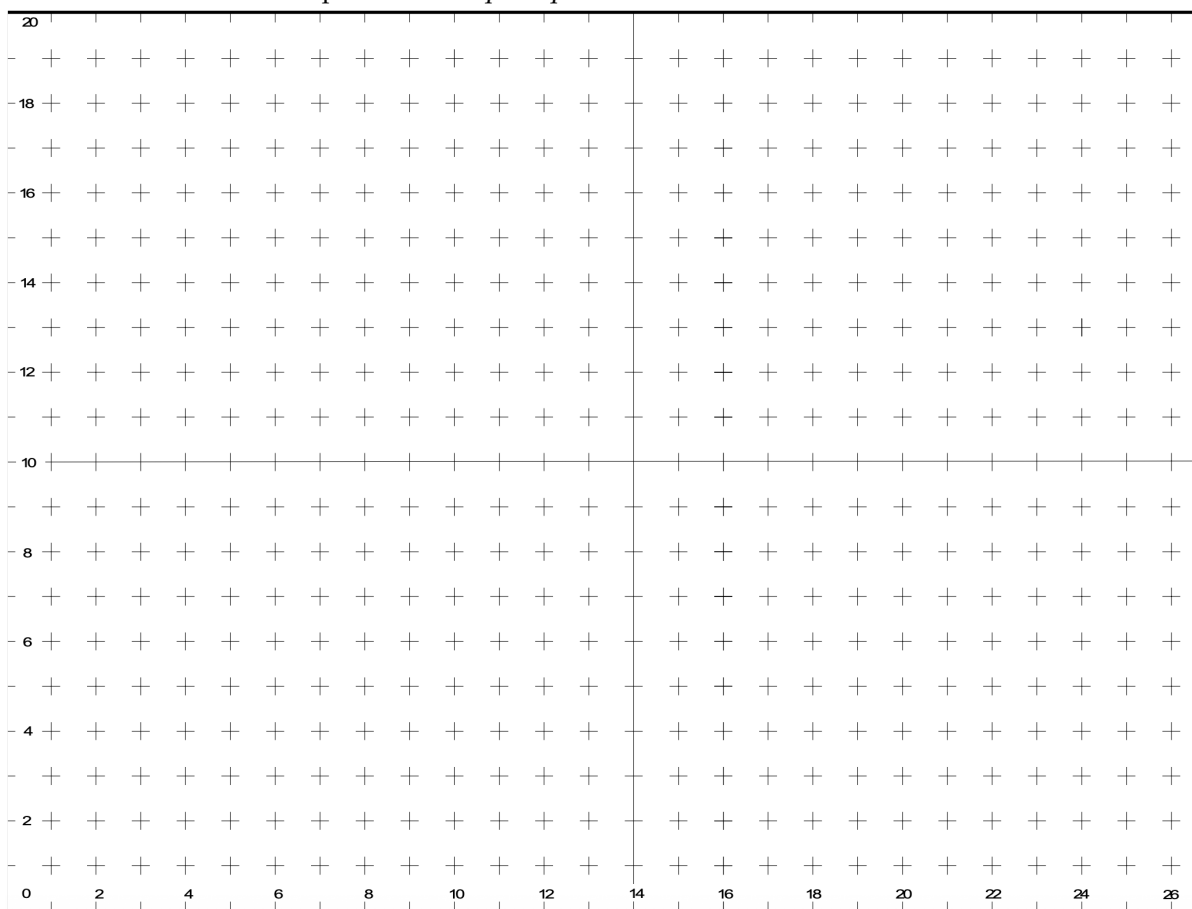
Question 12**1/2 points**

In which region(s) of the two-point pattern would the electric force on a test charge be largest? Why?

Question 13**3 points**

In Graph 2, sketch the electric field lines according to your measured mapping of equipotential surfaces for the parallel-plate electrode arrangement. Include arrows showing the direction of the field lines.

Graph 2: *Parallel-plate potential and electric field lines.*



+1+1+1

Question 14 **$\frac{1}{2}$ points**

Describe the trajectory of a positive unit charge introduced midway between the parallel plates with a velocity directed parallel to the plates. Draw a figure if you wish.